

HYPER-3

Which of three daily doses should go into phase III, and is any of them harming the people taking it?

Abstract

The study addresses which of three daily doses should go into phase III and whether any of them is harming the people taking it. The effect of arm on change is identified via the backdoor route. Four hundred adults with elevated or stage-1 hypertension were randomized to standard of care or to one of three daily doses, stratified by age band. Seated systolic pressure was measured weekly, with endpoints averaging readings inside windows. Each dose was compared with control by ANCOVA on the change from baseline, adjusting for baseline pressure and for age band when pooled. For the pooled analysis, the 40 mg dose compared with control was -1.53 mmHg (90% WALD -2.97 to -0.10 mmHg), the 20 mg dose compared with control was -4.80 mmHg (90% WALD -6.17 to -3.44 mmHg), and the 10 mg dose compared with control was -5.27 mmHg (90% WALD -6.63 to -3.91 mmHg). Subgroup analysis of the 40 mg dose compared with control yielded 6.20 mmHg (90% WALD 3.98 to 8.42 mmHg) for age 51+ and -5.26 mmHg (90% WALD -8.17 to -2.36 mmHg) for age 25-35. Twelve harm contrasts were monitored at scheduled safety reviews. The strongest standing assumptions are that dropout is ignorable given baseline and that the pooled contrast answers the question.

Reported quantities: 40 mg vs control, pooled; 20 mg vs control, pooled; 10 mg vs control, pooled; 40 mg vs control, age 51+

1. Introduction

This study evaluates which of three daily doses should advance to phase III and assesses whether any of the doses harms the participants taking it. The inquiry centers on the pooled contrast comparing the 40 mg dose against control. The decision criterion turns on a reference threshold of 0.00 mmHg. An interval estimate that lies entirely on one side of 0.00 mmHg settles the question. An interval that spans 0.00 mmHg leaves the question unsettled. In addition to average dose effects, the assessment requires determining whether harm occurs across the studied groups. The conclusions of the study are conditional throughout.

Specifically, the findings depend on two unresolved assumptions: that dropout is ignorable given baseline and that the pooled contrast answers the question. These assumptions govern the interpretation of both the comparative efficacy and safety evaluations.

2. Methods

The target of estimation is the effect of treatment arm on change in blood pressure, identified via the backdoor route. In the trial protocol, 400 adults with elevated or stage-1 hypertension were randomized 2:1:1:1 to standard of care or to one of three daily doses, stratified by age band. The design estimates three contrasts against a shared control rather than evaluating active doses against one another. Seated systolic pressure was measured weekly. Both study endpoints average readings across a window—weeks 5–8 for safety and weeks 9–12 for the primary endpoint—rather than relying on a single visit. Averaging correlated readings reduces the variance of an individual endpoint, enabling the safety boundary to operate within strata. Each dose was compared with control using ANCOVA on the change from baseline, adjusting for baseline pressure, and for age band when the contrast was pooled across bands. Randomization licenses the intent-to-treat contrast. Adjusting for pre-randomization covariates provides precision without altering the identification established by randomization. Twelve harm contrasts, evaluating each dose against control overall and within each age band, were monitored at scheduled safety reviews against a posterior-probability boundary with a 2 mmHg clinical margin. Pre-specifying the age bands as monitored contrasts permits harm within a specific stratum to be identified prior to study completion.

Assumptions this analysis rests on

Assumption	Statement	Status	Challenged by
dropout is ignorable given baseline	units who left the study differ from those who stayed only through what was measured before randomization	is assumed and has not been checked	a tipping-point analysis over plausible departures from ignorability
the pooled contrast answers the question	a single number over all age bands describes the effect the decision is about	is known to fail	heterogeneity across the pre-specified strata

Table 1. Standing assumptions

assumption Adherence was not adjusted for: it lies on the path from arm to outcome, and conditioning on it would break the randomization.

assumes *no_conditioning_on_post_randomization_variables* (satisfied): the analysis conditions on nothing measured after assignment; adherence is a mediator and dropout is a descendant of the outcome

assumption The pooled 40 mg contrast is reported alongside the per-band contrasts, never instead of them.

assumes *the_pooled_contrast_answers_the_question* (violated): a single number over all age bands describes the effect the decision is about

Assumption ledger

Step	Parameter	Value
Identification	route	backdoor
Identification	status	identified
Protocol	allocation	2:1:1:1
Protocol	arms	standard_of_care, dose_10, dose_20, dose_40

Protocol	enrollment weeks	16
Protocol	follow-up weeks	24
Protocol	primary endpoint week	12
Protocol	strata	age_25_35, age_36_50, age_51_plus
Protocol	units randomized	400
Measurement	minimum visits in window	3
Measurement	retention at final week	79.8%
Analysis	estimator	ANCOVA (OLS on change, baseline-adjusted)
Analysis	interval mass	90%
Analysis	pooled covariates	baseline_centred, is_age_36_50, is_age_51_plus
Analysis	within-stratum covariates	baseline_centred
Safety monitoring	clinical margin (mmHg)	2
Safety monitoring	monitored contrasts	12

Table 2. The design, as recorded parameters

3. Results

The pooled estimate for 40 mg versus control is -1.53 mmHg (90% Wald interval, -2.97 to -0.10 mmHg). For participants aged 51 and older, the estimated contrast for 40 mg versus control is 6.20 mmHg (90% Wald interval, 3.98 to 8.42 mmHg).

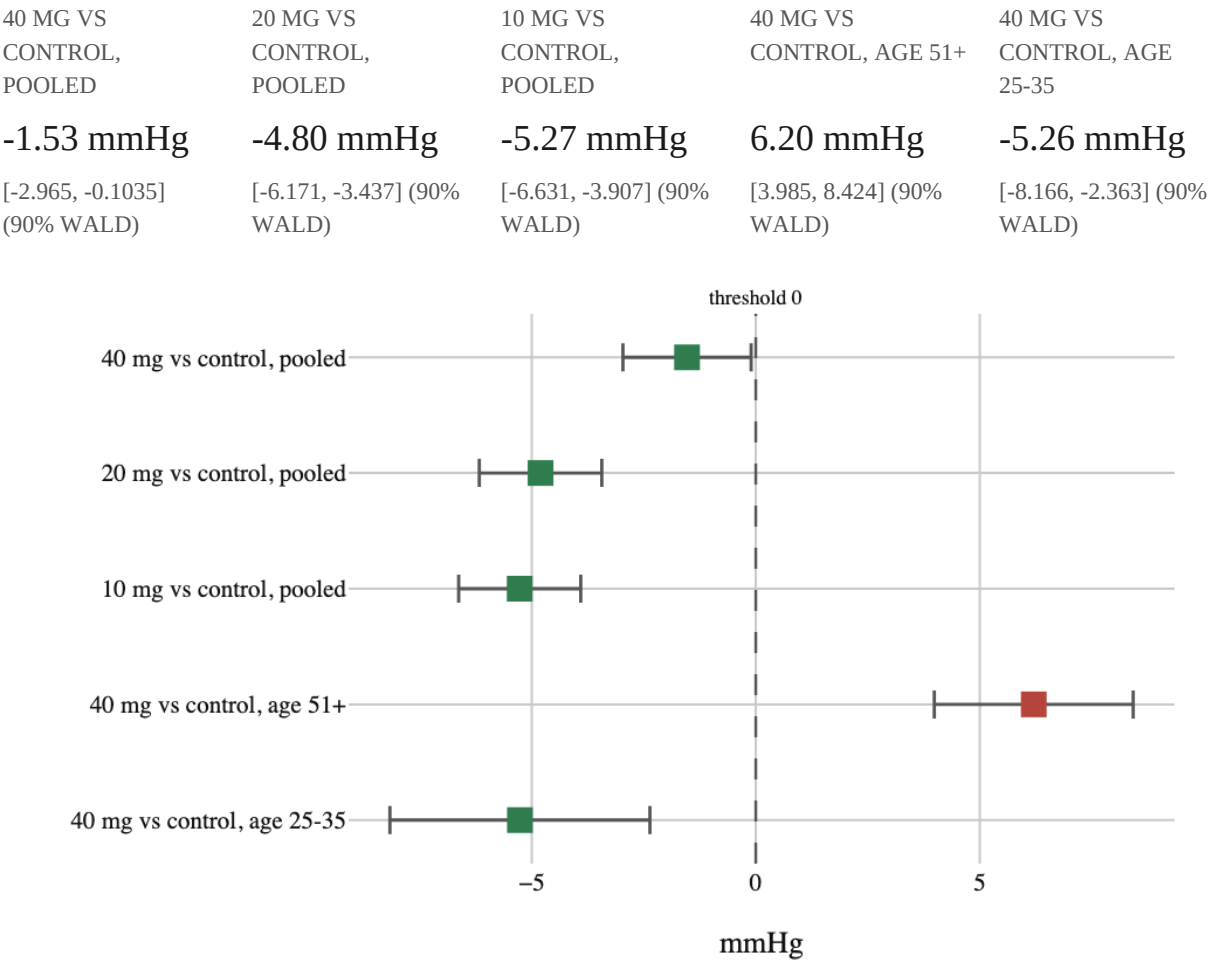


Figure 1. Estimated quantities with their intervals

Quantity	Estimate	Unit	Interval	Threshold	Relative to threshold
40 mg vs control, pooled	-1.53	mmHg	-2.97 to -0.10 (90% WALD)	0.00	below threshold
20 mg vs control, pooled	-4.80	mmHg	-6.17 to -3.44 (90% WALD)	0.00	below threshold
10 mg vs control, pooled	-5.27	mmHg	-6.63 to -3.91 (90% WALD)	0.00	below threshold
40 mg vs control, age 51+	6.20	mmHg	3.98 to 8.42 (90% WALD)	0.00	above threshold
40 mg vs control, age 25-35	-5.26	mmHg	-8.17 to -2.36 (90% WALD)	0.00	below threshold

Table 3. Estimated quantities

4. Model checking

Diagnostic checks describe model fit and retention rather than the treatment effect itself. Units still on study at the final week were 0.797. There were 372 units contributing to the primary endpoint. There were 100 units randomized in the age 25 to 35 band, 160 units randomized in the age 36 to 50 band, and 140 units randomized in the age 51 plus band. These diagnostics monitor sample availability and stratum allocations across follow-up. Diagnostic checks can reduce confidence in the estimates, but they never establish them. Evaluating model fit does not verify the unresolved assumptions underlying the estimation. In particular, that dropout is ignorable given baseline remains an unresolved assumption. That the pooled contrast answers the question is likewise an unresolved assumption. Retention metrics and diagnostic numbers cannot verify whether dropout is ignorable given baseline. Similarly, checking the fit does not demonstrate that the pooled contrast answers the question. These checks evaluate the operational sample without validating the unmeasured identification conditions.

UNITS STILL ON STUDY AT THE FINAL WEEK	UNITS CONTRIBUTING TO THE PRIMARY ENDPOINT	UNITS RANDOMIZED IN BAND AGE 25 35	UNITS RANDOMIZED IN BAND AGE 36 50	UNITS RANDOMIZED IN BAND AGE 51 PLUS
0.797	372	100	160	140

Check	Value	Unit	Interval
Units still on study at the final week	0.797	—	—
Units contributing to the primary endpoint	372	—	—
Units randomized in band age 25 35	100	—	—
Units randomized in band age 36 50	160	—	—
Units randomized in band age 51 plus	140	—	—

Table 4. Recorded diagnostics

5. Discussion

The effect is identified via the backdoor route, which licenses interpreting these estimates as treatment effects rather than associations under the trial assumptions. For the 10 mg dose versus control, the pooled effect is -5.27 mmHg (90% WALD -6.63 to -3.91 mmHg), where the interval lies entirely below the 0.00 mmHg threshold on the favourable side. For the 20 mg dose versus control, the pooled effect is -4.80 mmHg (90% WALD -6.17 to -3.44 mmHg), with the interval also falling entirely below the 0.00 mmHg threshold on the favourable side. For the 40 mg dose versus control, the pooled effect is -1.53 mmHg (90% WALD -2.97 to -0.10 mmHg), with an interval located entirely below 0.00 mmHg on the favourable side. This pooled estimate for the 40 mg dose should not be evaluated alone, as it averages across an age band that benefits and an age band that is harmed. For participants in the age band 25-35, the 40 mg dose versus control yields an effect of -5.26 mmHg (90% WALD -8.17 to -2.36 mmHg), lying entirely below the 0.00 mmHg threshold on the favourable side. For participants in the age band 51+, the 40 mg dose versus control yields an effect of 6.20 mmHg (90% WALD 3.98 to 8.42 mmHg). Because this interval lies entirely above 0.00 mmHg on the unfavourable side, it reflects the harm contrast that the pooled figure conceals and that the safety boundary was designed to monitor. All of these conclusions remain conditional on two unresolved assumptions: that dropout is ignorable given baseline, and that the pooled contrast answers the question.

6. Conclusions

The selection of a dose for phase III and the assessment of harm cannot be resolved with a single pooled figure, conditional on dropout being ignorable given baseline and the pooled contrast answering the question. For the 40 mg dose versus control among participants aged 51 and older, the estimated effect is 6.20 mmHg (90% WALD 3.98 to 8.42 mmHg), which sits on the unfavorable side of the threshold. Four other findings sit on the favorable side of the threshold. The 40 mg dose versus control in the age 25 to 35 band is -5.26 mmHg (90% WALD -8.17 to -2.36 mmHg). Across pooled age bands, the 10 mg dose versus control is -5.27 mmHg (90% WALD -6.63 to -3.91 mmHg). The pooled 20 mg dose versus control is -4.80 mmHg (90% WALD -6.17 to -3.44 mmHg). The pooled 40 mg dose versus control is -1.53 mmHg (90% WALD -2.97 to -0.10 mmHg). A single pooled summary averages harm in one age stratum with benefit in another. Consequently, the pooled figure is not a substitute for the stratum-specific estimates, and the disaggregated findings form the substantive result. The effect is identified via the backdoor route. These conclusions hold under the unresolved assumptions that dropout is ignorable given baseline and that the pooled contrast answers the question.

7. Limitations

Two unresolved assumptions perform inferential work that the data alone cannot settle. Stating each assumption explicitly identifies what it licenses and what would challenge its validity. The first assumption is that dropout is ignorable given baseline measurements. Under this condition, units who left the study are assumed to differ from those who remained only through characteristics recorded prior to randomization. This assumption licenses evaluating treatment effects from the observed outcomes conditional on baseline adjustments. The ignorability of dropout remains an assumption and has not been verified. It would be challenged by a tipping-point analysis evaluating plausible departures from ignorability. The second assumption is that the pooled contrast answers the decision question. This assumes that a single summary measure across all age bands adequately describes the effect under evaluation. It licenses relying on the aggregate contrast rather than stratum-specific estimates to guide dose selection. This assumption is known to fail. It is challenged by treatment effect heterogeneity across the pre-specified age strata.

8. Provenance

Every number in this document resolves to a quantity in the evidence record below. Prose was checked against that record: any numeral not traceable to it, and any claim the record does not itself make, is reported rather than published.

Quantity	Key	Value	Produced by
40 mg vs control, pooled	pooled_40	-1.53 mmHg (90% WALD -2.97 to -0.10 mmHg)	identify.ols (ANCOVA)
20 mg vs control, pooled	pooled_20	-4.80 mmHg (90% WALD -6.17 to -3.44 mmHg)	identify.ols (ANCOVA)
10 mg vs control, pooled	pooled_10	-5.27 mmHg (90% WALD -6.63 to -3.91 mmHg)	identify.ols (ANCOVA)
40 mg vs control, age 51+	oldest_40	6.20 mmHg (90% WALD 3.98 to 8.42 mmHg)	identify.ols (ANCOVA)
40 mg vs control, age 25-35	youngest_40	-5.26 mmHg (90% WALD -8.17 to -2.36 mmHg)	identify.ols (ANCOVA)
Units still on study at the final week	retention	0.797	—
Units contributing to the primary endpoint	analysed	372	—
Units randomized in band age 25 35	n_age_25_35	100	—
Units randomized in band age 36 50	n_age_36_50	160	—
Units randomized in band age 51 plus	n_age_51_plus	140	—

Table 5. Quantities and their sources

Field	Value
endpoint	mean change, weeks 9-12
interval_mass	90%
seed	20260821
world	nbs/case-studies/hypertension/hyper3.py
evidence hash	962d1a7970f3f2dfdc0297e7d9d6d16dd568a5441fc1cfaa7c44837d40f9c2e5
narration: abstract	gemini-3.5-flash-lite — verified
narration: introduction	gemini-3.7-flash — verified
narration: methods	gemini-3.7-flash — verified
narration: results	gemini-3.7-flash — verified

narration: diagnostics	gemini-3.7-flash — verified
narration: discussion	gemini-3.7-flash — verified
narration: conclusions	gemini-3.7-flash — verified
narration: limitations	gemini-3.7-flash — verified

Table 6. Run